

FutureProof EPM

Income from Property Equity — Technical Architecture

A comprehensive guide to the platform architecture, agents, workflows, and compliance systems that power monthly income distribution for equity-partnered homeowners.

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1. EPM Overview

FutureProof's Income from Property Equity solution is a fund-based income product that transforms home equity into sustainable monthly income. The model connects four stakeholders through a sophisticated distribution system powered by monthly rebalancing and performance monitoring.

The Four Stakeholders

Stakeholder	Role	Capital Flow
Customer	Homeowner who partners equity for monthly income	Receives monthly distributions from fund returns
Investment Partner	Manages invested capital	Generates returns; manages fund performance and rebalancing
Lender	Provides mortgage product & customer relationship	Receives margin on monthly distributions
Wholesale Funder	Supplies capital to lenders via funding lines	Receives wholesale margin on deployed capital

Monthly Distribution Flow

Fund Returns: Investment Partner's holdings generate returns that are aggregated and tracked monthly (on average 10% p.a. over 30 years).

Monthly Distribution: System calculates monthly payout amounts based on:

- Customer's equity percentage (typically 2% of home value)
- Fund performance month-to-date
- Interest holiday status (if triggered)

Amount to Customer: Customer receives their share of monthly returns as income.

Lender Margin: Lender deducts their margin from distribution (0.75% to 2% depending on tier).

Wholesale Margin: Wholesale Funder receives their cut (0.2% to 3.6% depending on structure).

FutureProof Margin: Platform takes its cut (0.25%) for ongoing management.

2. Customer Journey

PHASE 1: ORIGINATION (Weeks 1–4)

Customer applies: Provides property details, equity percentage, income expectations through portal.

Deal Agent processes: KYC/AML checks, property valuation, risk tier assignment, document generation.

e-Signature: Customer reviews and signs loan documents, investment agreement.

Funding confirmation: Lender & Wholesale Funder approve. Capital transferred. Account activated.

PHASE 2: ONGOING SERVICING (Months 1–30+)

Monthly distributions: Distribution Agent calculates returns every month. Customer receives payment on fixed date.

Performance monitoring: Investment Agent tracks fund performance. Generates rolling metrics visible on customer dashboard.

Interest holidays: If performance drops, holiday flag automatically triggered. Holiday status visible on customer statements and dashboard.

Statements & reports: Monthly statements show distribution amount, margins paid, fund performance, and holiday status.

Account management: Customer can request early exit, modify settings. Lender reviews requests within 5 business days.

PHASE 3: EXIT (Early or Maturity)

Customer-initiated: Request early repayment. Deal Agent calculates exit position (fund value minus outstanding fees).

Lender approval: Lender reviews exit request, confirms repayment terms.

Settlement: Distribution Agent processes final payment. All accounts reconciled. Loan closed.

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Documents archived: All transaction history, statements, audit logs preserved for 7 years.

3. Stakeholder Account Views

Key information visible to each participant

Stakeholder	Key Account Views	Key Actions	Fraud Detection
Customer	• Monthly distribution amount • YTD cumulative income • Fund performance chart • Interest holiday status	• View transaction history • Download statements • Manage account settings • Request early exit	• Multiple withdrawals same day • Unusual location access • Rapid device changes • Profile mismatch
Lender	• Monthly distribution schedule • Margin received • Customer payment status • Portfolio performance	• Approve interest holidays • View customer profiles • Adjust risk tier • Monitor portfolio health	• Systematic holiday requests • Unusual pattern changes • Distribution delays • Sudden customer exits
Investment Partner	• Fund performance metrics • ETF correlation data • Rebalancing schedule • Cost basis tracking	• Execute rebalancing • Manage ETF holdings • Report performance • Adjust hedges	• Unexplained underperformance • ETF holdings mismatches • Rebalancing delays • Cost basis discrepancies
Wholesale Funder	• Portfolio yield metrics • Default & recovery rates • Margin collections • Lender tier performance	• Approve lender tiers • Adjust funding lines • Initiate risk reviews • Approve new partners	• Yield compression • Concentration risk • Lender financial strain • Default clustering

4. System Architecture

Core Platform

Rails 8.1 + Hotwire + Stimulus: Single monolithic Rails application with real-time updates via Hotwire (Turbo) for seamless user interactions and Stimulus controllers for client-side interactivity. Hotwire enables live dashboard updates (fund performance, distribution status, holiday triggers) without requiring WebSocket complexity.

PostgreSQL Database: Single source of truth for all customer, transaction, and account data with full audit trails. Uses JSON columns for flexible fund metadata storage, materialized views for performance metrics aggregation, and database triggers to enforce compliance rules automatically. Partitioned by customer ID for horizontal scaling.

Kafka Event Queue: Asynchronous event streaming for agent coordination and system-wide notifications. Agents subscribe to topics: `distributions.*`, `holidays.*`, `compliance.*`, `deals.*`. Ensures exactly-once processing with idempotent subscriber logic; dead-letter queues capture processing failures.

Redis Cache: Session management, rate limiting, and temporary computation caching. Used for: customer session state, monthly distribution calculations (staged before final commit), Bloom filters for fraud detection, cache-aside pattern for frequently accessed data.

S3 Storage: Document archival, statements, and audit logs with versioning enabled. Uses server-side encryption (SSE-S3). Lifecycle policies archive logs after 90 days to Glacier, delete after 7 years (compliance requirement).

Core Services

Distribution Engine: Microservice (Ruby + Sidekiq) that runs on 1st-5th of month. Pulls monthly fund returns from Investment Partner, calculates per-customer distributions. Stages calculations in Redis for audit, executes ACH/bank transfers via Stripe, publishes `distribution.executed` event. Implements retry logic with exponential backoff.

Compliance Engine: Real-time service that validates every transaction against 20+ regulatory rules. Writes audit trail to PostgreSQL with immutable event log table. Implements Change Data Capture (CDC) via database triggers.

Fraud Detection: Machine learning pipeline trained on historical transaction patterns. Real-time scorer checks each transaction against: (1) known fraud patterns (Bloom filter), (2) behavioral anomalies (isolation forest), (3) rule-based checks (impossible geography, velocity limits).

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Interest Holiday Logic: Scheduled job runs monthly. Queries fund performance history, calculates rolling 12-month returns, compares against entry/exit thresholds. State machine manages transitions.

Notification Service: Async job queue with multiple channels: email (SendGrid), SMS (Twilio), in-app notifications. Implements delivery guarantees: stores pending notifications in database, retries with backoff, marks as delivered only after confirmation.

Document Engine: Server-side rendered (ERB templates) + Prawn gem. Generates: loan documents, investment agreements, monthly statements, disclosures. All outputs digitally signed, archived to S3 with immutable version IDs.

5. Four-Agent Architecture

Autonomous agents managing workflows and monitoring

Agent	Primary Role	Key Workflows	Triggers
Deal Agent	Customer acquisition & onboarding	• Application processing • KYC/AML verification • Document generation & e-sign • Loan approval	• New application • Compliance clearance • Fund confirmation
Distribution Agent	Monthly returns & payment	• Calculate distributions • Deduct margins • Execute payments • Generate statements	• Monthly cutoff date • Fund performance updates • Holiday status changes
Investment Agent	Fund monitoring & holidays	• Monitor fund performance • Calculate rolling metrics • Determine holiday eligibility • Trigger/release holidays	• Monthly performance updates • Rolling threshold breach • Recovery confirmation
Compliance Agent	Monitoring & enforcement	• Real-time fraud detection • Regulatory rule enforcement • Generate audit reports • Alert suspicious activity	• Transaction submitted • Fraud score threshold • Rule violation detected

Agent Communication Flow

All agents publish events to Kafka; others subscribe to relevant events and react asynchronously. This decouples the four workflows and enables real-time coordination without tight coupling.

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Deal Agent publishes deal.created when customer onboarded. Distribution Agent and Compliance Agent subscribe to initiate workflows.

Distribution Agent publishes distribution.monthly on cutoff date. Lender and Wholesale Funder receive payments.

Investment Agent publishes holiday.triggered or holiday.released. Distribution Agent adjusts payment calculations.

Compliance Agent publishes fraud.detected or rule.violation. System escalates to lender/admin for review.

6. Agent Technology Implementation

Agent Framework

Four agents implemented as Rails service classes (e.g., DealAgent::OnboardingService). Each agent runs as a Sidekiq background job (worker pool, 4 workers per pod). Agents are horizontally scalable.

Event Subscription (Kafka)

Each agent subscribes to relevant Kafka topics using ruby-kafka gem. Consumer groups enable automatic partitioning. Dead-letter queues capture processing failures; operations team investigates manually.

Idempotency & Retry Logic

Agents implement idempotent handlers. Uses database unique constraints + transactional locks. Failed agent tasks auto-retry with exponential backoff (1s → 2s → 4s → 8s → 16s), then escalate to dead-letter queue.

Distributed Tracing

Every event gets trace_id (UUID). As event flows through agents, each writes logs tagged with trace_id. Enables end-to-end visibility: search CloudWatch by trace_id to see entire customer journey.

State Management

Deal workflow: draft → submitted → kyc_pending → approved → funded → active

Holiday logic: normal → holiday_pending → holiday_active → recovery_in_progress → normal

Call Center Integration (Twilio Flex)

When human escalation needed, ops team receives ticket in Twilio Flex dashboard. Agent clicks customer, Flex dials phone via Twilio, logs call recording automatically. If customer calls in, Flex IVR routes to available agent. Agent sees customer profile, recent interactions, account balance.

Synchronous vs. Asynchronous

Agent actions default to async (publish event, job queued). Synchronous calls used only for: (a) Stripe payment processing (5-second timeout), (b) IDology KYC check (10-second timeout), (c) property valuation lookup (15-second timeout). All wrapped in circuit breaker pattern.

7. Interest Holiday Mechanism

Dynamic flag based on fund performance

How Holidays Work

Interest holidays provide borrowers protection during market downturns. The Investment Agent monitors rolling fund performance against dynamic thresholds.

Parameter	Value	Meaning
Entry Threshold	0.90 (90%)	If fund value drops to 90% of starting value, entry conditions met
Exit Threshold	1.458 (145.8%)	When fund recovers to 145.8% of original value, holiday automatically released
Maximum Duration	Variable	Holiday automatically expires if conditions not met within timeframe
Accumulated Interest	Ongoing	During holiday, interest is not distributed to customer but accumulates in account

Timeline Example

Months 1–10: Normal Operations - Fund performing well. Customer receives monthly distributions. Investment Agent monitors thresholds monthly.

Month 11: Market Downturn - Fund value drops to 88% of starting value. Entry threshold (90%) breached. Investment Agent flags interest holiday status in customer account.

Months 12–18: Holiday Flagged - Interest holiday flag remains active. Customer continues receiving monthly distributions. Lender monitoring alert active. Monthly statements indicate holiday status.

Month 19: Recovery Confirmed - Fund recovers to 145.8% threshold. Investment Agent removes holiday flag. Status returns to normal operations.

8. Compliance & Fraud Detection

Regulatory adherence and system safeguards

Compliance Rules

Rule	Validation	Failure Action
KYC/AML Verification	Customer identity verified before onboarding	Application rejected; customer notified
Responsible Lending	Fund amount appropriate for customer profile	Risk tier adjusted; margin increased
Distribution Accuracy	Monthly calculations verified; margins correct	Distribution held pending audit
Fund Performance Audit	Daily returns match investment partner reports	Discrepancy flagged; reconciliation initiated
Holiday Logic	Thresholds correctly applied; payment suspension verified	Holiday manually reviewed
Data Privacy	PII access logged; encryption verified	Unauthorized access alert; investigation

Real-Time Fraud Monitoring

Account Access Anomalies: Logins from unusual locations, rapid device changes, after-hours patterns

Unusual Transaction Patterns: Multiple withdrawals same day, requests after distribution, systematic holiday manipulation

Data Inconsistencies: Mismatches between reported and actual fund values, distribution calculation errors

Regulatory Violations: KYC/AML failures, undisclosed conflicts, unauthorized transfers

Audit & Reporting

Audit Trail: Every transaction, margin payment, and holiday trigger recorded with timestamp, user ID, change details. Logs retained 7 years minimum.

Monthly Reporting: Lenders and wholesale funders receive automated reports on portfolio health, margin collections, default scenarios.

Annual Compliance Certification: External auditor review. Covers all regulatory requirements across Australia, UK, and US jurisdictions.

9. Technology Stack

Backend & Processing

- Rails 8.1 (Ruby 3.4.8)
- Sidekiq 7.x (background jobs, 4 workers)
- ruby-kafka gem (event streaming)
- Puma web server (3 threads, auto-scaling)

Data & Persistence

- PostgreSQL 16 (primary + read replica)
- Redis 7 (caching + session store)
- Kafka 3.6 (event log, 3 replicas)
- PgBouncer (connection pooling)

Agent & Workflow Technology

- Sidekiq (async job execution)
- Kafka (event pub/sub, 4 topic streams)
- OpenTelemetry (distributed tracing, trace ID tagging)
- Dead-letter queues (failure handling)
- State machines (holiday logic, deal workflow)

Customer Communication

- SendGrid (transactional email)
- Twilio (SMS + voice calls)
- Twilio Flex (call center platform)
- WebRTC (in-app video calls)
- DocuSign (e-signature, document workflows)

Frontend & UI

- Hotwire (Turbo Streams + Drive)

- Stimulus JS (lightweight controllers)
- Tailwind CSS (utility-first styling)
- ERB templates (server-rendered)
- Chart.js (performance visualization)

Storage & Infrastructure

- AWS S3 (documents, versioned)
- AWS CloudFront (CDN)
- S3 Glacier (archive)
- Fly.io (deployment)
- GitHub Actions (CI/CD)
- Docker (containerization)
- Terraform (IaC)

Monitoring & Security

- Sentry (error tracking)
- CloudWatch (logs)
- Grafana (dashboards)
- PagerDuty (escalation)
- HTTPS/TLS, WAF, CSP headers

10. Database Schema

Core Tables

Table	Purpose	Key Fields
customers	Homeowners enrolled in partnership	id, email, phone, kyc_status, aml_status, risk_tier
properties	Residential with equity partnerships	id, customer_id, address, property_value, equity_percentage, loan_amount
accounts	Customer investment accounts	id, customer_id, property_id, starting_balance, current_balance, holiday_flagged
fund_performance	Monthly fund metrics (materialized view)	account_id, month, starting_value, ending_value, return_pct, rolling_12m_return
distributions	Monthly customer distributions (immutable)	id, account_id, fund_return_amount, lender_margin, customer_payout, executed_at
payments	ACH/bank transfer records (immutable)	id, distribution_id, amount, bank_account_id, status, stripe_charge_id
audit_log	Immutable event log (append-only)	id, entity_type, action, changes_json, user_id, timestamp, trace_id
fraud_alerts	Detected suspicious patterns	id, customer_id, alert_type, fraud_score, status, reviewed_by
lenders	Loan originators & servicers	id, name, tier, max_funding_line, current_balance, margin_rate
wholesale_funders	Capital providers to lenders	id, name, funding_capacity, current_deployed, margin_rate

Key Design Patterns

Immutable Logs: distributions, payments, and audit_log tables have no UPDATE/DELETE permissions. All changes recorded as INSERT-only events with timestamps. Ensures regulatory compliance and audit trail integrity.

Soft Deletes: customers and accounts include deleted_at column (nullable). Logical deletion preserves historical data; queries filter WHERE deleted_at IS NULL.

JSON Columns: audit_log.changes_json stores before/after values. fraud_alerts.details_json stores detection model output. accounts.metadata_json stores region-specific terms.

Materialized Views: fund_performance refreshed nightly (expensive aggregation of historical returns). Improves dashboard load times.

Partitioning: distributions and payments partitioned by created_at (monthly partitions). Improves query performance for range scans. Older partitions moved to cold storage.

11. Deployment & Quality Assurance

Testing Strategy

Unit Tests: Core business logic (distribution calculations, holiday logic, fraud detection rules).

Integration Tests: Agent workflows, end-to-end customer journeys, payment processing.

Performance Tests: Monthly distribution batch processing at scale (1000s of customers), query optimization.

Security Tests: Authentication, authorization, data access controls, PII encryption.

Manual Testing: Customer portal UX, lender reporting accuracy, admin workflows.

Continuous Deployment

GitHub Actions: Automated tests run on every PR. RSpec (unit/integration, 200+ specs), Rubocop (linting), Brakeman (security), database migrations validated. Tests must pass + code review approval before merge.

Staging Environment: Deployed automatically from main branch using Fly.io staging app (same infra as production). Runs against production-replica database (encrypted, daily sanitized PII). Manual smoke tests performed.

Production Release: Blue-green deployment on Fly.io. Traffic gradually shifted 10% → 50% → 100% to new version over 10 minutes (canary rollout). Health checks verify: Rails boot, database connection, Kafka topics, endpoint latency. Automatic rollback if health checks fail within 5 minutes.

Monitoring & Alerting

Sentry: Real-time error tracking with source map support. Critical errors trigger PagerDuty escalations within 2 minutes.

CloudWatch: Structured JSON logging (via Rails lograge). Each request logs: trace_id, user_id, duration_ms, database_queries, external_api_calls. Retention: 90 days hot, archival to S3 after.

Grafana Dashboards: Real-time visibility into: (1) System health (CPU, memory, disk usage), (2) Application metrics (requests/sec, error rate, p95 latency), (3) Business metrics (distributions executed, payments, revenue), (4) Agent health (processing lag, dead-letter queue size), (5) Database health (connection pool, replication lag). Alerts: error rate >1%, distribution latency >60s, Kafka lag >1000 messages.

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